



Easy. Accurate. Available. Lung Function Testing

# HL7 CONFORMANCE STATEMENT

**PRODUCT DETAILS:**

MiniBox+

**MANUFACTURER:**

PulmOne Advanced Medical Devices Ltd.

14 HaMelacha street

Ra'anana, 4366107 ISRAEL

US Tel: 1 844 PulmOne (785 6663)

Tel: +972 (77) 510 0938

info@pulm-one.com

**STANDARDS:**

HL7 standard v2

MiniBox+™ is a registered trademark of PulmOne Advanced Medical Devices Ltd.

Copyright © 2024 PulmOne Ltd. All rights reserved.

|                                                                                                                                                       |                                                     |                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------|
| <div>  <br/> Easy. Accurate. Available. Lung Function Testing </div> |                                                     |                                             |
| Date :<br>12 Sept 2024                                                                                                                                | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130<br>Revision: 01 |
| Page   2                                                                                                                                              |                                                     |                                             |

Table of Contents

Introduction ..... 3

    Overview ..... 3

    Intended Audience..... 3

    Standard Version Control..... 3

Overview ..... 4

    Diagram..... 4

    General..... 5

    Logical Architecture ..... 5

    PulmOne MiniBox Tablet Configuration ..... 6

    Security Architecture ..... 6

    HL7 Interface Description and requirements ..... 6

Detailed HL7 messages and segments description..... 7

    Order Message (ORMO01)..... 7

        Message Header (MSH) ..... 7

        Patient Id (PID)..... 8

        Patient Visit (PV1) ..... 9

        Common Order (ORC) ..... 11

        Observation Request (OBR) ..... 12

        Sample ORMO01 message from an HL7 EMR server to PulmOne MiniBox ..... 13

    Observation Result Unsolicited (ORUR01)..... 14

        Message Header (MSH) ..... 14

        Patient Id (PID)..... 15

        Patient Visit (PV1) ..... 16

        Common Order (ORC) ..... 18

        Common Order (OBR) ..... 19

        Observation Result (OBX)..... 20

        Sample ORUR01 sent from the MiniBox to the EMR HL7 results interface..... 22

|                                                                                                                                       |                                                     |                             |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------|
| <br>Easy. Accurate. Available. Lung Function Testing |                                                     |                             |
| Date :<br>12 Sept 2024                                                                                                                | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   3                                                                                                                              |                                                     | Revision: 01                |

## Introduction

### Overview

This document provides the requirements to interoperate with the PulmOne MiniBox+ using the Health Level Seven (HL7) standard.

The objective of this document is to provide a standard set of HIS/CIS/RIS requirements to effectively support and connect to the PulmOne MiniBox+.

### Intended Audience

The user of this document is involved with system integration and/or software design. We assume that the reader is familiar with the terminology and concepts that are used in HL7 V2 standard. Readers not familiar with HL7 V2 terminology should first read the appropriate parts of the HL7 V2 standard itself, prior to reading this conformance statement.

Although the use of this conformance statement in conjunction with the HL7 V2 standard is intended to facilitate communication with PulmOne MiniBox+, it is not sufficient to guarantee, by itself the inter-operation of the connection between PulmOne MiniBox+ and the 3rd party HL7-based system.

The integration of any device into a system of interconnected devices goes beyond the scope of the HL7 V2 standard and this conformance statement when interoperability is desired. The responsibility for analyzing the applications requirements and developing a solution that integrates the PulmOne MiniBox+ with other vendors' systems is the user's responsibility.

### Standard Version Control

The PulmOne MiniBox+ supports version 2.5.1 of the HL7 V2 protocol. The MiniBox+ has also been successfully integrated with HL7 servers using HL7 V2 versions 2.3 and 2.4. In general, the PulmOne MiniBox+ is capable of integrating with servers using higher versions of HL7 V2. In such cases contact the PulmOne service team for further advice.



|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   5               |                                                     | Revision: 01                |

## General

- The PulmOne MiniBox device is capable of working in either a standalone mode or alternatively, it can function as an HL7 client, communicating with an HL7 EMR server, using a subset of the HL7 protocol as describe in Figure 1.
- All the software and hardware required for the HL7 communications are built into the MiniBox tablet.
- The MiniBox receives the patients and visits information from the EMR server using the HL7 **ORMO01** message. The MiniBox supports both new visits as well as visit cancellations.
- The MiniBox sends the test results back to the EMR server using the HL7 **ORUR01** message. This message includes the measured values of all the discrete parameters that were selected by the user using the settings screen, as well as their predicted values, their lower and upper level of normal and their Z-Score.
- The ORUR01 message sent by the MiniBox also includes an encapsulated visit summary PDF report, encoded using Base64. Alternatively, the MiniBox can store the summary PDF report on a network drive, provided by the customer and include the name of the visit's summary PDF report file as part of the ORUR01 message.

## Logical Architecture

PulmOne's MiniBox device enables Physicians to conduct pulmonary and diffusion testing. The device includes a measuring hardware unit (12 x 18 x 20 in) and a 12in tablet which runs the minibox software that controls the measuring unit. The software is implemented using Windows 10 operating system, .NET Framework, C#, WPF, MVVM.

- Presentation Layer: Includes a custom UI layer that the physician and patients interact with, using the tablet's 12 in touchscreen that includes a virtual keyboard.
- Application Layer: Communicates with the Measurement hardware unit, controls the measurements and validates the measurement results.
- Data Access & Integration Layer: Communicates with the EMR server using the HL7 protocol. Receives the patient demographic and required testing from the EMR server and sends the test results, as well as the Physician diagnostics to the EMR server.
- Data Layer: Stores the Measurement results on a local SQLite database that resides on the MiniBox tablet.

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   6               |                                                     | Revision: 01                |

## PulmOne MiniBox Tablet Configuration

| PulmOne MiniBox Tablet Specifications                          |
|----------------------------------------------------------------|
| Tablet model – DELL Latitude 5175 or Microsoft Go 3            |
| CPU: Intel® Core™ m5-6Y57 Processor (4M Cache, up to 2.80 GHz) |
| Memory: 4GB (1x4GB) DDR3L 1600MHz Memory                       |
| Storage: M.2 128GB SATA Class 20 Solid State Drive             |
| Operating System – Windows 10 Professional                     |

- No external Servers required
- Alternatively using a Jumper EZPAD PRO 6 tablet with similar specifications

## Security Architecture

- The PulmOne MiniBox includes functionality for encryption/decryption of all confidential patient and test results data. The data is saved in an SQLite database that resides on an encrypted drive in the tablet itself.
  - Local Encryption method: Windows BitLocker
- The PulmOne MiniBox supports authorization module by user name and password.
  - The system enforces password rules when creating passwords (min length, at least one of each lower/upper case letters, digits and special characters).
  - The system locks the account after 5 login failures.
  - The system logs the user out automatically after 10 min of inactivity (configurable) and returns to the home screen.

## HL7 Interface Description and requirements

- Connection Type: TCP/IP
- MiniBox Listening Port: 8080 (configurable)
- The Firewall (if any) needs to allow TCP/IP connection/messages from the IP address of the EMR server HL7 order interface to the MiniBox listening port.
- The Firewall (if any) needs to allow connections/messages from the IP address of the MiniBox to the EMR server results interface listening port.
- It is expected that the DHCP server assigns a static IP address for the MiniBox (and the EMR server) to prevent connections/messages being blocked after a power failure or network maintenance.

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   7               |                                                     | Revision: 01                |

## Detailed HL7 messages and segments description

### Order Message (ORMO01)

The HL7 Order Message (ORMO01) from the HL7 EMR server to the PulmOne MiniBox should contain the following segments:

- Message Header (MSH)
- Patient Id (PID)
- Patient Visit (PV1)
- Common Order (ORC)
- Observation Request (OBR)

Fields marked in the following tables as Required, are parsed and used. Fields Marked as Optional, are parsed and echoed back in the ORUR01 message. Fields marked as Not Used are ignored.

#### Message Header (MSH)

| Message Header (MSH) |                                  |           |                 |
|----------------------|----------------------------------|-----------|-----------------|
| Position             | Element                          | Required? | Sample Content  |
| 0                    | Message Name                     | Required  | MSH             |
| 1                    | Field Separator                  | Required  |                 |
| 2                    | Encoding Characters              | Required  | ^~\&            |
| 3                    | Sending Application              | Required  | EMR Server Name |
| 4                    | Sending Facility                 | Required  | Facility Name   |
| 5                    | Receiving Application            | Optional  | MiniBox         |
| 6                    | Receiving Facility               | Optional  | PulmOne         |
| 7                    | Date Time of message             | Required  | 20201105133018  |
| 8                    | Security                         | Not Used  |                 |
| 9                    | Message Type                     | Required  | ORM^O01         |
| 10                   | Message Control                  | Optional  | P               |
| 11                   | Processing Id                    | Not Used  |                 |
| 12                   | Version Id                       | Not Used  |                 |
| 13 to 18             |                                  | Not Used  |                 |
| 19                   | Principle Language of Message(*) | Optional  | SPIRO           |

(\*) Starting in version 3.3.0 of the MiniBox+ application, Not Used in previous versions

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   8               |                                                     | Revision: 01                |

#### Patient Id (PID)

| Patient Identifier (PID) |                        |           |                                   |
|--------------------------|------------------------|-----------|-----------------------------------|
| Position                 | Element                | Required? | Sample Content                    |
| 0                        | Message Name           | Required  | PID                               |
| 1                        | Set Id                 | Not Used  |                                   |
| 2                        | Patient Id-External    | Not Used  |                                   |
| 3                        | Patient Id-Internal    | Required  | 52068866                          |
| 4                        | Alternate Patient Id   | Not Used  |                                   |
| 5                        | Patient Name           | Required  | LASTNAME^FIRSTNAME                |
| 6                        | Mother Maiden Name     | Not Used  |                                   |
| 7                        | Date of Birth          | Required  | 19760821                          |
| 8                        | Sex                    | Required  | M or F                            |
| 9                        | Patient Alias          | Not Used  |                                   |
| 10                       | Ethnic Group           | Optional  | Caucasian                         |
| 11                       | Patient Address        | Optional  | 123 FAKE ST^^NEW YORK^NY^10010^US |
| 12                       | County Code            | Not Used  |                                   |
| 13                       | Phone Number-Home      | Optional  | (212)555-5555                     |
| 14                       | Phone Number-Business  | Not Used  |                                   |
| 15                       | Language               | Not Used  |                                   |
| 16                       | Marital Status         | Not Used  |                                   |
| 17                       | Religion               | Not Used  |                                   |
| 18                       | Patient Account Number | Optional  | 496677136                         |
| 19                       | Social Security Number | Optional  | 123-45-6789                       |
| 20                       | Driver's License       | Not Used  |                                   |
| >=21                     |                        | Not Used  |                                   |



|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   9               |                                                     | Revision: 01                |

## Patient Visit (PV1)

| Patient Visit (PV1) |                           |           |                |
|---------------------|---------------------------|-----------|----------------|
| Position            | Element                   | Required? | Sample Content |
| 0                   | Message Name              | Required  | PV1            |
| 1                   | Set Id                    | Not Used  |                |
| 2                   | Patient Class             | Not Used  |                |
| 3                   | Assigned Patient Location | Not Used  |                |
| 4                   | Admission Type            | Not Used  |                |
| 5                   | Pre-Admit Number          | Not Used  |                |
| 6                   | Prior Patient Location    | Not Used  |                |
| 7                   | Attending Doctor          | Optional  | FAKE^NAME1     |
| 8                   | Referring Doctor          | Optional  | FAKE^NAME2     |
| 9                   | Consulting Doctor         | Not Used  |                |
| 10                  | Hospital Service          | Not Used  |                |
| 11                  | Temporary Location        | Not Used  |                |
| 12                  | Pre-Admit Test Indicator  | Not Used  |                |
| 13                  | Re Admission Indicator    | Not Used  |                |
| 14                  | Admit Source              | Not Used  |                |
| 15                  | Ambulatory Status         | Not Used  |                |
| 16                  | VIP Indicators            | Not Used  |                |
| 17                  | Admitting Doctor          | Not Used  |                |
| 18                  | Patient Type              | Not Used  |                |
| 19                  | Visit Number              | Optional  | 22228615       |
| 20                  | Financial Class           | Not Used  |                |
| 21                  | Charge Price Indicator    | Not Used  |                |
| 22                  | Courtesy Code             | Not Used  |                |
| 23                  | Credit Rating             | Not Used  |                |
| 24                  | Contract Code             | Not Used  |                |
| 25                  | Contract Effective Date   | Not Used  |                |
| 26                  | Contract Amount           | Not Used  |                |
| 27                  | Contract Period           | Not Used  |                |
| 28                  | Interest Code             | Not Used  |                |
| 29                  | Transfer To Bad Debt Code | Not Used  |                |
| 30                  | Transfer To Bad Debt Date | Not Used  |                |
| 31                  | Bad Debt Agency Code      | Not Used  |                |
| 32                  | Bad Debt Transfer Amount  | Not Used  |                |
| 33                  | Bad Debt Recovery Amount  | Not Used  |                |

Date :  
12 Sept 2024

Page | 10

## MiniBox+ HL7 Conformance Statement

Document #:  
PUO-DOC20130

Revision: 01

|      |                          |          |  |
|------|--------------------------|----------|--|
| 34   | Delete Account Indicator | Not Used |  |
| 35   | Delete Account Date      | Not Used |  |
| 36   | Discharge Disposition    | Not Used |  |
| 37   | Discharge To Location    | Not Used |  |
| 38   | Diet Type                | Not Used |  |
| 39   | Servicing Facility       | Not Used |  |
| 40   | Bed Status               | Not Used |  |
| 41   | Account Status           | Not Used |  |
| 42   | Pending Location         | Not Used |  |
| 43   | Prior Temporary Location | Not Used |  |
| 44   | Admit Date Time          | Not Used |  |
| 45   | Discharge Date Time      | Not Used |  |
| 46   | Current Patient Balance  | Not Used |  |
| 47   | Total Charges            | Not Used |  |
| 48   | Total Adjustments        | Not Used |  |
| 49   | Total Payments           | Not Used |  |
| >=50 |                          | Not Used |  |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   11              |                                                     | Revision: 01                |

## Common Order (ORC)

| Common Order (ORC) |                          |           |                                           |
|--------------------|--------------------------|-----------|-------------------------------------------|
| Position           | Element                  | Required? | Sample Content                            |
| 0                  | Message Name             | Required  | ORC                                       |
| 1                  | Order Control            | Required  | NW (new order) or CA (order cancellation) |
| 2                  | Placer Order Number      | Not Used  |                                           |
| 3                  | Filler Order Number      | Not Used  |                                           |
| 4                  | Placer Group Number      | Not Used  |                                           |
| 5                  | Order Status             | Not Used  |                                           |
| 6                  | Response Flag            | Not Used  |                                           |
| 7                  | Timing Quantity          | Not Used  |                                           |
| 8                  | Parent                   | Not Used  |                                           |
| 9                  | Date Time of Transaction | Not Used  |                                           |
| 10                 | Entered Ny               | Not Used  |                                           |
| 11                 | Verified By              | Not Used  |                                           |
| 12                 | Ordering Provider        | Not Used  |                                           |
| >=13               |                          | Not Used  |                                           |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   12              |                                                     | Revision: 01                |

### Observation Request (OBR)

| Observation Request (OBR) |                                 |           |                               |
|---------------------------|---------------------------------|-----------|-------------------------------|
| Position                  | Element                         | Required? | Sample Content                |
| 0                         | Message Name                    | Required  | OBR                           |
| 1                         | Set Id                          | Not Used  |                               |
| 2                         | Placer Order Number             | Optional  | 73289718                      |
| 3                         | Filler Order Number             | Not Used  |                               |
| 4                         | Universal Service Id            | Required  | PFT13^PULMONARY FUNCTION TEST |
| 5                         | Priority                        | Not Used  |                               |
| 6                         | Requested Date Time             | Not Used  |                               |
| 7                         | Observation Date Time           | Not Used  |                               |
| 8                         | Observation End date Time       | Not Used  |                               |
| 9                         | Collection Volume               | Not Used  |                               |
| 10                        | Collection Id                   | Not Used  |                               |
| 11                        | Specimen Action Code            | Not Used  |                               |
| 12                        | Danger Code                     | Not Used  |                               |
| 13                        | Relevant Clinical Info          | Not Used  |                               |
| 14                        | Specimen Received Date Time     | Not Used  |                               |
| 15                        | Specimen Source                 | Not Used  |                               |
| 16                        | Ordering Provider               | Optional  | NAME                          |
| 17                        | Order Callback Phone number     | Not Used  |                               |
| 18                        | Placers Field 1                 | Not Used  |                               |
| 19                        | Placers Field 2                 | Not Used  |                               |
| 20                        | Filler Field 1                  | Not Used  |                               |
| 21                        | Filler Field 2                  | Not Used  |                               |
| 22                        | Results Report Status Change dt | Not Used  |                               |
| 23                        | Charge To Practice              | Not Used  |                               |
| 24                        | Diagnostic Service Sect Id      | Not Used  |                               |
| 25                        | Result Status                   | Not Used  |                               |
| 26                        | Linked Results                  | Not Used  |                               |
| 27                        | Quantity timing                 | Not Used  |                               |
| 28                        | Results Copies To               | Not Used  |                               |
| 29                        | Parent Accession Number         | Not Used  |                               |
| 30                        | Transportation Mode             | Not Used  |                               |
| 31                        | Reason for Study                | Not Used  |                               |
| 32                        | Print Result Interpreter        | Not Used  |                               |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   13              |                                                     | Revision: 01                |

|      |                           |          |  |
|------|---------------------------|----------|--|
| 33   | Assist Result Interpreter | Not Used |  |
| 34   | Technician                | Not Used |  |
| 35   | Transcriptionist          | Not Used |  |
| 36   | Scheduled Date Time       | Not Used |  |
| >=37 |                           | Not Used |  |

Other valid options for OBR4:

PulmOneSPR^Pulm One Spirometry → for FVC only

PulmOneSVC^Pulm One Slow Vital → for SVC only

PulmOneMVV^Pulm One Voluntary → for MVV only

PFT13^COMPLETE PULMONARY FUNCTION TEST → FVC, LVM and DLCO

PFT13^PULMONARY FUNCTION TEST → FVC, LVM and DLCO

PFT Full → FVC, LVM and DLCO

Sample ORM001 message from an HL7 EMR server to PulmOne MiniBox

```
MSH|^~\&|EPIC|FakeFacility|||20201105133018||ORM^O01|22|T|2.3
PID|||52068865||LASTNAME^FIRST ||19760821|F|||123 FAKE ST^^NEW
YORK^NY^10010^US^^^60|| (212) 555-5555||||496677135
PV1|||||||||||||||||||||||||||||||||||||||||
ORC|NW|162406141|||||20201105133000^R||20201105133017|FAKE^
^JOSSIE^^||
OBR||162406141||PFT13^COMPLETE PULMONARY FUNCTION
TEST||20201105133000|20201105133017|||||||
```

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   14              |                                                     | Revision: 01                |

## Observation Result Unsolicited (ORUR01)

The HL7 Observation Result Unsolicited (ORUR01) sent from the PulmOne MiniBox to the HL7 EMR server would contain the following segments:

- Message Header (MSH)
- Patient Id (PID)
- Patient Visit (PV1)
- Common Order (ORC)
- Observation Request (OBR)
- Observation Result (OBX)

Fields marked as Required in the following tables are always included. Fields Marked as Optional are parsed from the ORUR01 message and echoed back. Fields marked as Not Used are not included in the message.

### Message Header (MSH)

| Message Header (MSH) |                              |           |                 |
|----------------------|------------------------------|-----------|-----------------|
| Position             | Element                      | Required? | Sample Content  |
| 0                    | Message Name                 | Required  | MSH             |
| 1                    | Field Separator              | Required  |                 |
| 2                    | Encoding Characters          | Required  | ^^\&            |
| 3                    | Sending Application          | Required  | EMR Server Name |
| 4                    | Sending Facility             | Required  | Facility Name   |
| 5                    | Receiving Application        | Required  | MiniBox         |
| 6                    | Receiving Facility           | Required  | PulmOne         |
| 7                    | Date Time of message         | Required  | 20201105133018  |
| 8                    | Security                     | Not Used  |                 |
| 9                    | Message Type                 | Required  | ORU^R01         |
| 10                   | Message Control              | Optional  | P               |
| 11                   | Processing Id                | Not Used  |                 |
| 12                   | Version Id                   | Not Used  |                 |
| 13                   | Sequence Number              | Not Used  |                 |
| 14                   | Continuation Pointer         | Not Used  |                 |
| 15                   | Accept Ack Type              | Not Used  |                 |
| 16                   | Application Ack type         | Not Used  |                 |
| 17                   | Country Code                 | Not Used  |                 |
| 18                   | Character Set                | Not Used  |                 |
| 19                   | Principal Language of Msg(*) | Optional  | SPIRO           |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   15              |                                                     | Revision: 01                |

|      |                            |          |                                                  |
|------|----------------------------|----------|--------------------------------------------------|
| 20   | Alt Character Set Handling | Not Used |                                                  |
| 21   | Msg Profile Id             | Required | LRI_NG_RN_Profile^HL7^2.16.840.1.113883.9.20^ISO |
| >=22 |                            | Not Used |                                                  |

(\*) Starting in version 3.3.0 of the MiniBox+ application, Not Used in previous versions

#### Patient Id (PID)

| Patient Identifier (PID) |                        |           |                    |
|--------------------------|------------------------|-----------|--------------------|
| Position                 | Element                | Required? | Sample Content     |
| 0                        | Message Name           | Required  | PID                |
| 1                        | Set Id                 | Not Used  |                    |
| 2                        | Patient Id-External    | Not Used  |                    |
| 3                        | Patient Id-Internal    | Required  | 52068866           |
| 4                        | Alternate Patient Id   | Not Used  |                    |
| 5                        | Patient Name           | Required  | LASTNAME^FIRSTNAME |
| 6                        | Mother Maiden Name     | Not Used  |                    |
| 7                        | Date of Birth          | Required  | 19760821           |
| 8                        | Sex                    | Required  | F                  |
| 9                        | Patient Alias          | Not Used  |                    |
| 10                       | Ethnic Group           | Required  | Caucasian          |
| 11                       | Patient Address        | Not Used  |                    |
| 12                       | County Code            | Not Used  |                    |
| 13                       | Phone Number-Home      | Not Used  |                    |
| 14                       | Phone Number-Business  | Not Used  |                    |
| 15                       | Language               | Not Used  |                    |
| 16                       | Marital Status         | Not Used  |                    |
| 17                       | Religion               | Not Used  |                    |
| 18                       | Patient Account Number | Optional  | 496677136          |
| 19                       | Social Security Number | Not Used  |                    |
| 20                       | Driver's License       | Not Used  |                    |
| >=21                     |                        | Not Used  |                    |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   16              |                                                     | Revision: 01                |

#### Patient Visit (PV1)

| Patient Visit (PV1) |                           |           |                |
|---------------------|---------------------------|-----------|----------------|
| Position            | Element                   | Required? | Sample Content |
| 0                   | Message Name              | Required  | PV1            |
| 1                   | Set Id                    | Not Used  |                |
| 2                   | Patient Class             | Not Used  |                |
| 3                   | Assigned Patient Location | Not Used  |                |
| 4                   | Admission Type            | Not Used  |                |
| 5                   | Pre-Admit Number          | Not Used  |                |
| 6                   | Prior Patient Location    | Not Used  |                |
| 7                   | Attending Doctor          | Optional  | FAKE^NAME1     |
| 8                   | Referring Doctor          | Optional  | FAKE^NAME2     |
| 9                   | Consulting Doctor         | Not Used  |                |
| 10                  | Hospital Service          | Not Used  |                |
| 11                  | Temporary Location        | Not Used  |                |
| 12                  | Pre-Admit Test Indicator  | Not Used  |                |
| 13                  | Re Admission Indicator    | Not Used  |                |
| 14                  | Admit Source              | Not Used  |                |
| 15                  | Ambulatory Status         | Not Used  |                |
| 16                  | VIP Indicators            | Not Used  |                |
| 17                  | Admitting Doctor          | Not Used  |                |
| 18                  | Patient Type              | Not Used  |                |
| 19                  | Visit Number              | Optional  | 22228615       |
| 20                  | Financial Class           | Not Used  |                |
| 21                  | Charge Price Indicator    | Not Used  |                |
| 22                  | Courtesy Code             | Not Used  |                |
| 23                  | Credit Rating             | Not Used  |                |
| 24                  | Contract Code             | Not Used  |                |
| 25                  | Contract Effective Date   | Not Used  |                |
| 26                  | Contract Amount           | Not Used  |                |
| 27                  | Contract Period           | Not Used  |                |
| 28                  | Interest Code             | Not Used  |                |
| 29                  | Transfer To Bad Debt Code | Not Used  |                |
| 30                  | Transfer To Bad Debt Date | Not Used  |                |
| 31                  | Bad Debt Agency Code      | Not Used  |                |
| 32                  | Bad Debt Transfer Amount  | Not Used  |                |
| 33                  | Bad Debt Recovery Amount  | Not Used  |                |



Date :  
12 Sept 2024

Page | 17

## MiniBox+ HL7 Conformance Statement

Document #:  
PUO-DOC20130

Revision: 01

|      |                          |          |  |
|------|--------------------------|----------|--|
| 34   | Delete Account Indicator | Not Used |  |
| 35   | Delete Account Date      | Not Used |  |
| 36   | Discharge Disposition    | Not Used |  |
| 37   | Discharge To Location    | Not Used |  |
| 38   | Diet Type                | Not Used |  |
| 39   | Servicing Facility       | Not Used |  |
| 40   | Bed Status               | Not Used |  |
| 41   | Account Status           | Not Used |  |
| 42   | Pending Location         | Not Used |  |
| 43   | Prior Temporary Location | Not Used |  |
| 44   | Admit Date Time          | Not Used |  |
| 45   | Discharge Date Time      | Not Used |  |
| 46   | Current Patient Balance  | Not Used |  |
| 47   | Total Charges            | Not Used |  |
| 48   | Total Adjustments        | Not Used |  |
| 49   | Total Payments           | Not Used |  |
| >=50 |                          | Not Used |  |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   18              |                                                     | Revision: 01                |

#### Common Order (ORC)

| Common Order (ORC) |                           |           |                           |
|--------------------|---------------------------|-----------|---------------------------|
| Position           | Element                   | Required? | Sample Content            |
| 0                  | Message Name              | Required  | ORC                       |
| 1                  | Order Control             | Required  | RE                        |
| 2                  | Placer Order Number       | Optional  | 12345                     |
| 3                  | Filler Order Number       | Required  | OBR^5_01122019_021633^FVC |
| 4                  | Placer Group Number       | Not Used  |                           |
| 5                  | Order Status              | Not Used  |                           |
| 6                  | Response Flag             | Not Used  |                           |
| 7                  | Timing Quantity           | Not Used  |                           |
| 8                  | Parent                    | Not Used  |                           |
| 9                  | Date Time of Transaction  | Not Used  |                           |
| 10                 | Entered By                | Not Used  |                           |
| 11                 | Verified By               | Not Used  |                           |
| 12                 | Ordering Provider         | Optional  | Name                      |
| 13                 | Enterers Location         | Not Used  |                           |
| 14                 | Callback Phone Number     | Not Used  |                           |
| 15                 | Order Effective Date Time | Not Used  |                           |
| 16                 | Order Control Code Reason | Not Used  |                           |
| 17                 | Entering Organization     | Not Used  |                           |
| 18                 | Entering Device           | Not Used  |                           |
| 19                 | Action By                 | Not Used  |                           |
| >=20               |                           | Not Used  |                           |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   19              |                                                     | Revision: 01                |

### Common Order (OBR)

| Observation Request (OBR) |                                 |            |                               |
|---------------------------|---------------------------------|------------|-------------------------------|
| Position                  | Element                         | Required ? | Sample Content                |
| 0                         | Message Name                    | Required   | OBR                           |
| 1                         | Set Id                          | Required   | 1                             |
| 2                         | Placer Order Number             | Optional   | 73289718                      |
| 3                         | Filler Order Number             | Required   | OBR^5_01122019_021633^FVC     |
| 4                         | Universal Service Id            | Required   | PFT13^PULMONARY FUNCTION TEST |
| 5                         | Priority                        | Not Used   |                               |
| 6                         | Requested Date Time             | Not Used   |                               |
| 7                         | Observation Date Time           | Required   | 20201105133018                |
| 8                         | Observation End date Time       | Not Used   |                               |
| 9                         | Collection Volume               | Not Used   |                               |
| 10                        | Collection Id                   | Not Used   |                               |
| 11                        | Specimen Action Code            | Not Used   |                               |
| 12                        | Danger Code                     | Not Used   |                               |
| 13                        | Relevant Clinical Info          | Not Used   |                               |
| 14                        | Specimen Received Date Time     | Not Used   |                               |
| 15                        | Specimen Source                 | Not Used   |                               |
| 16                        | Ordering Provider               | Optional   | NAME                          |
| 17                        | Order Callback Phone number     | Not Used   |                               |
| 18                        | Placers Field 1                 | Not Used   |                               |
| 19                        | Placers Field 2                 | Not Used   |                               |
| 20                        | Filler Field 1                  | Not Used   |                               |
| 21                        | Filler Field 2                  | Not Used   |                               |
| 22                        | Results Report Status Change dt | Not Used   |                               |
| 23                        | Charge To Practice              | Not Used   |                               |
| 24                        | Diagnostic Service Sect Id      | Required   | PF                            |
| 25                        | Result Status                   | Required   | F                             |
| 26                        | Linked Results                  | Not Used   |                               |
| 27                        | Quantity timing                 | Not Used   |                               |
| 28                        | Results Copies To               | Not Used   |                               |
| 29                        | Parent Accession Number         | Not Used   |                               |
| 30                        | Transportation Mode             | Not Used   |                               |
| 31                        | Reason for Study                | Not Used   |                               |
| 32                        | Print Result Interpreter        | Not Used   |                               |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   20              |                                                     | Revision: 01                |

|      |                           |          |  |
|------|---------------------------|----------|--|
| 33   | Assist Result Interpreter | Not Used |  |
| 34   | Technician                | Not Used |  |
| 35   | Transcriptionist          | Not Used |  |
| 36   | Scheduled Date Time       | Not Used |  |
| >=37 |                           | Not Used |  |

#### Observation Result (OBX)

| Observation Result (OBX) |                              |           |                          |
|--------------------------|------------------------------|-----------|--------------------------|
| Position                 | Element                      | Required? | Sample Content           |
| 0                        | Message Name                 | Required  | OBX                      |
| 1                        | Set Id                       | Required  | 1                        |
| 2                        | Value Type                   | Required  | NM or ST or ED           |
| 3                        | Observation Identifier       | Required  | 19868-9_Best^FVC_Best^LN |
| 4                        | Observation Sub Id           | Not Used  |                          |
| 5                        | Observation Result           | Required  | 2.53                     |
| 6                        | Units                        | Required  | L                        |
| 7                        | Reference Range              | Not Used  |                          |
| 8                        | Abnormal Flags               | Not Used  |                          |
| 9                        | Probability                  | Not Used  |                          |
| 10                       | Nature Of Abnormal Test      | Not Used  |                          |
| 11                       | Observation Result Status    | Required  | F                        |
| 12                       | Date Last Normal Value       | Not Used  |                          |
| 13                       | User Defined Access Checks   | Not Used  |                          |
| 14                       | Date Time of The Observation | Not Used  |                          |
| 15                       | Procedure's Id               | Not Used  |                          |
| 16                       | Responsible Observer         | Not Used  |                          |
| 17                       | Observation Method           | Not Used  |                          |
| >=18                     |                              | Not Used  |                          |

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   21              |                                                     | Revision: 01                |

For each measured parameter, that was selected by the user to be included in the summary PDF report in the parameter settings screen, the following six OBX segments are included:

1. Best measured value
2. Predicted value
3. Percent of the measured value out of the predicted value
4. Lower Limit of Normal (LLN) for that parameter
5. Upper Limit of Normal (ULN) for that parameter
6. Z-score for that parameter (the number of standard deviations between the measured value and the predicted value)

```
OBX|1|ST|19868-9_Best^FVC_Best^LN||5.03|L||||F|||||
OBX|2|ST|19868-9_Pred^FVC_Pred^LN||4.43|L||||F|||||
OBX|3|ST|19868-9_%Pred^FVC_%Pred^LN||113.45|%||||F|||||
OBX|4|ST|19868-9_LLN^FVC_LLN^LN||3.42|L||||F|||||
OBX|5|ST|19868-9_ULN^FVC_ULN^LN||5.46|L||||F|||||
OBX|6|ST|19868-9_ZSCORE^FVC_ZSCORE^LN||0.96|None||||F|||||
```

For tests that support both Pre and Post Bronchodilator measurements, there will be 9 OBX segment per parameter:

1. Best measured value Pre-Bronchodilator
2. Predicted Result
3. Percent of the Pre-Bronchodilator measured value out of the predicted value
4. Lower Limit of Normal (LLN) for that parameter
5. Upper Limit of Normal (ULN) for that parameter
6. Z-score for that Pre-Bronchodilator measured value (the number of standard deviations between the measured value and the predicted value)
7. Best measured value Post-Bronchodilator
8. Percent of the Post-Bronchodilator measured value out of the predicted value
9. Z-score for that Post-Bronchodilator measured value (the number of standard deviations between the Post-Bronchodilator measured value and the predicted value)

```
OBX|1|ST|19868-9_Best_Pre^FVC_Best_Pre^LN||5.03|L||||F|||||
OBX|2|ST|19868-9_Pred^FVC_Pred^LN||4.43|L||||F|||||
OBX|3|ST|19868-9_%Pred_Pre^FVC_%Pred_Pre^LN||113.45|%||||F|||||
```

|                        |                                  |                             |
|------------------------|----------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b>                  | Document #:<br>PUO-DOC20130 |
| Page   22              | <b>HL7 Conformance Statement</b> | Revision: 01                |

```
OBX|4|ST|19868-9_LLN^FVC_LLN^LN||3.42|L||||F|||||
OBX|5|ST|19868-9_ULN^FVC_ULN^LN||5.46|L||||F|||||
OBX|6|ST|19868-9_ZSCORE_Pre^FVC_ZSCORE_Pre^LN||0.96|None||||F|||||
OBX|7|ST|19868-9_Best_Post^FVC_Best_Post^LN||5.03|L||||F|||||
OBX|8|ST|19868-9_%Pred_Post^FVC_%Pred_Post^LN||113.45|%||||F|||||
OBX|9|ST|19868-9_ZSCORE_Post^FVC_ZSCORE_Post^LN||0.96|None||||F|||||
```

The format of the OBR segment for the encapsulated visit summary pdf report:

```
OBX|283|ED|58477-1^Pulmonary Function Test Report^VSPULR||^AP^PDF^Base64^.....
```

The maximum length of the pre encoded data in one encapsulated pdf OBX segment was fixed at 30000 bytes for all PulmOne MiniBox versions prior to version 3.1.3. The encapsulated summary PDF report is usually longer than 30000 bytes, so it was split into multiple OBX segments, and each segment was individually Base64 encoded. The receiving application had to Base64 decode the Base64 encoded content of each encapsulated pdf OBR segment before appending them to recreate the PDF report.

Starting from version 3.1.3 of the PulmOne Minibox, the maximum number of bytes in each OBR segment can be increased using a property in the MiniBox config file, reducing the number of OBR segments required to transmit the encapsulated pdf report.

## Sample ORUR01 sent from the MiniBox to the EMR HL7 results interface

```
MSH|^~\&|MiniBox|PulmOne|||202110311513|0|ORU^R01|19_24102021_19
0349||252||||||LRI_NG_RN_Profile^HL7^2.16.840.1.113883.9.20^I
SO
```

```
PID|||1470||LastName^FirstName||19760827|Male||Caucasian||||||
||
```

```
PV1||||||||||||||||||||||||||||||||||||||||
```

```
ORC|RE||OBR^19_24102021_190349^FVC||||||||||||
```

```
OBR|1||OBR^19_24102021_190349^FVC|FVC|||20211024190349||||||
|||||PF|F||||||||
```

```
OBX|1|ST|19868-9_Best^FVC_Best^LN||5.03|L||||F|||||
```

|                        |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| Date :<br>12 Sept 2024 | <b>MiniBox+</b><br><b>HL7 Conformance Statement</b> | Document #:<br>PUO-DOC20130 |
| Page   23              |                                                     | Revision: 01                |

OBX|2|ST|19868-9\_Pred^FVC\_Pred^LN||4.43|L|||||F|||||

OBX|3|ST|19868-9\_%Pred^FVC\_%Pred^LN||113.45|%|||||F|||||

OBX|4|ST|19868-9\_LLN^FVC\_LLN^LN||3.42|L|||||F|||||

OBX|5|ST|19868-9\_ULN^FVC\_ULN^LN||5.46|L|||||F|||||

OBX|6|ST|19868-9\_ZSCORE^FVC\_ZSCORE^LN||0.96|None|||||F|||||

OBX|7|ST|20150-9\_Best^FEV1\_Best^LN||3.92|L|||||F|||||

OBX|8|ST|20150-9\_Pred^FEV1\_Pred^LN||3.48|L|||||F|||||

OBX|9|ST|20150-9\_%Pred^FEV1\_%Pred^LN||112.77|%|||||F|||||

OBX|10|ST|20150-9\_LLN^FEV1\_LLN^LN||2.66|L|||||F|||||

OBX|11|ST|20150-9\_ULN^FEV1\_ULN^LN||4.25|L|||||F|||||

OBX|12|ST|20150-9\_ZSCORE^FEV1\_ZSCORE^LN||0.94|None|||||F|||||

OBX|13|ST|19926-5\_Best^FEV1/FVC\_Best^LN||77.99|%|||||F|||||

OBX|14|ST|19926-5\_Pred^FEV1/FVC\_Pred^LN||78.58|%|||||F|||||

OBX|15|ST|19926-5\_%Pred^FEV1/FVC\_%Pred^LN||99.25|%|||||F|||||

OBX|16|ST|19926-5\_LLN^FEV1/FVC\_LLN^LN||67.04|%|||||F|||||

OBX|17|ST|19926-5\_ULN^FEV1/FVC\_ULN^LN||88.56|%|||||F|||||

OBX|18|ST|19926-5\_ZSCORE^FEV1/FVC\_ZSCORE^LN||-  
0.09|None|||||F|||||

...